F70LA – Life Insurance Mathematics A, 2025/26
Chapter 4 – Insurance Benefits

Read Dickson, Hardy & Waters, Chapter 4.

We now bring together the concepts of present values (from financial mathematics) and our mathematics for death and survival functions to calculate "expected present values" for various insurance benefits, notably:

- Whole life insurance,
- Term insurance,
- Pure endowment,
- Endowment insurance,

and their variations.

4.1 Discounting a single payment in a life insurance context

We assume throughout that interest rates are constant. Let the annual effective rate of interest be $i > -1$ per annum with the corresponding annual discount factor is

$$v := \frac{1}{1+i}.$$

The force of interest is $\delta := \ln(1+i)$ per annum, and $v = e^{-\delta}$.

In this course, assume that references to an 'interest rate' mean an 'effective rate of interest' unless explicitly told otherwise.

A stream of future payments has a value today called the present value. A payment of:

- £1 paid in a constant $t \geq 0$ years' time has present value £v^t.
- £1 paid in a random $T \geq 0$ years' time has present value £v^T. Here, the present value is a random variable, being a deterministic function of the random variable T.

Let's set the random variable $T := T_x$, the random time of death of an individual (x), so that we value a payment of £1 paid at the time of death of (x). Then, as the pdf of T_x is ${}_tp_x\,\mu_{x+t}$, the expected present value of the payment is

$$\mathbb{E}(v^{T_x}) = \int_0^\infty v^t\,{}_tp_x\,\mu_{x+t}dt.$$

If the survival model is known (which means we know $\{{}_tp_x\}$ and $\{\mu_{x+t}\}$) and if we know the interest rate i, then $\mathbb{E}(v^{T_x})$ can be evaluated. If it cannot be calculated explicitly then it can be calculated numerically. Numerical integration is explored in F70LB Life Insurance Mathematics B.

4.2 Whole life insurance

A *whole life insurance* or *whole of life insurance* pays out a sum of money, called the *sum insured*, upon the death of the insured life. Whole life insurance is typically purchased by individuals who want lifelong coverage and a guaranteed payout to their beneficiaries upon their death. It's often chosen by those who want to leave money to their dependents or others or to cover potential inheritance tax.

The policy documents will specify:

- Who is the insured life and what is their date of birth. Writing (x) encapsulates this information in our theoretical context.
- For how long the policy lasts, i.e. its term. For a whole life insurance, the term is the lifetime of the policyholder.
- How much is the sum insured, i.e. how much is paid out upon the death of the insured life. The standard contract that we value pays out £1.
  - The sum insured is also called the *death benefit* in these type of insurances.
- When is the sum insured paid out after the insured life has died. Below, we consider how to value a contract when the sum insured is paid out: (i) immediately upon death and (ii) at the end of the year of death.

You may also see this policy called a whole life *assurance*. The word *assurance* may be used when the event which triggers payment (here, death whenever it occurs) is certain to happen within the term of the policy. As whole life policies last as long as the policyholder is alive, then death is certain to occur within the term of the whole life insurance policy. In this case, the *sum insured* can be called the *sum assured*.

4.2.1 Whole life insurance payable immediately on death

Definition 4.1 *A whole life insurance that pays a sum insured of £1 immediately upon the death of* (x) *has a random present value*

$$Z := v^{T_x} = e^{-\delta T_x}.$$

Its expected present value is represented in international actuarial notation as

$$\bar{A}_x := \mathbb{E}\left(v^{T_x}\right) = \int_0^\infty e^{-\delta t}\, {}_tp_x \mu_{x+t} dt.$$

Remark 4.2 *Why is the expected value useful? Assume that future lifetimes can be modelled by independent random variables. If an insurance company sells hundreds of whole life insurance policies to hundreds of lives, each paying out £1 if a life dies, then the observed payouts averaged across all the policies sold will be very close to the expected payout on a single policy. This is an application of the Law of Large Numbers. Allowing for different amounts of sum insured, then more policies need to be sold so that the average payout of a group of of policies with similar sum insureds is close to the expected payout on one of the policies.*

Example 4.3 *Suppose* (40) *buys a whole life insurance policy and suppose* $\bar{A}_{40} = 0.5$. *Then the single premium (i.e. one-off cost, with no further payments due from the policyholder) paid by* (40) *should be at least* £0.50 *for £1 of sum insured. The actual premium charged will be higher due to the insurance company's costs, profit, etc.*

The value of $\bar{A}_x$ depends on

- The interest rate, and
- The survival model.

We call these assumptions the *basis for the calculation* of $\bar{A}_x$, or the *basis*.

Understanding the integrand

Consider $\bar{A}_x = \int_0^\infty e^{-\delta t}\, {}_tp_x \mu_{x+t} dt$. We need to learn how to interpret integrands such as $e^{-\delta t}\, {}_tp_x \mu_{x+t}$.

Remember that $\mu_{x+t}\,\Delta$ is approximately equal to the probability of (x) dying in the time interval $[t, t+\Delta)$, given that they are alive at time t.

- Partition the entire time line $[0, \infty)$ into disjoint intervals $[0, \Delta), [\Delta, 2\Delta), [2\Delta, 3\Delta) \ldots$, for a small value of $\Delta > 0$.
- Take a representative interval, $[t, t+\Delta)$. Assume that the sum insured £1 is payable at the start of $[t, t+\Delta)$.
- Then the expected present value of the sum insured if (x) dies within $[t, t+\Delta)$ is

$$\underbrace{{}_tp_x}_{\substack{\text{Probability } (x) \text{ survives} \\ \text{to time } t.}} \times \underbrace{\mu_{x+t}\,\Delta}_{\substack{\text{Probability } (x) \text{ dies} \\ \text{in the time} \\ \text{interval } (t, t+\Delta), \\ \text{given they are} \\ \text{alive at time } t.}} \times \underbrace{1}_{\substack{\text{Amount of sum} \\ \text{insured paid} \\ \text{at time } t \text{ for} \\ \text{death occurring} \\ \text{in the time} \\ \text{interval } (t, t+dt).}} \times \underbrace{e^{-\delta t}}_{\substack{\text{Discount payment from} \\ \text{time } t \text{ to time } 0.}} .$$

- Sum the expression over all time intervals $[0, \Delta), [\Delta, 2\Delta), [2\Delta, 3\Delta) \ldots$, and then let $\Delta \downarrow 0$. In the limit, Δ becomes dt and the sum becomes an integral. Hence we obtain

$$\bar{A}_x = \underbrace{\int_0^\infty}_{\substack{\text{Sum over} \\ \text{all intervals} \\ (t, t+dt)}} \underbrace{{}_tp_x\, \mu_{x+t}\, dt}_{\substack{\text{Probability } (x) \text{ survives} \\ \text{to time } t \text{ and then} \\ \text{dies in the time} \\ \text{interval } (t, t+dt)}} \times \underbrace{1}_{\substack{\text{Amount of sum} \\ \text{insured paid} \\ \text{at time } t \text{ for} \\ \text{death occurring} \\ \text{in the time} \\ \text{interval } (t, t+dt).}} \times \underbrace{e^{-\delta t}}_{\substack{\text{Discount payment} \\ \text{from time } t \\ \text{to time 0.}}} .$$

Variance of the PV of the whole life insurance payable immediately on death

The present value random variable of a whole life insurance that pays £1 upon the death of (x) is $Z := e^{-\delta T_x}$. Its variance is $\mathbb{V}ar(Z) = \mathbb{E}(Z^2) - (\mathbb{E}(Z))^2$. The second moment has a special symbol, ${}^2\bar{A}_x$. It is defined as follows.

Definition 4.4 *The actuarial notation*

$${}^2\bar{A}_x := \mathbb{E}(Z^2) = \mathbb{E}\left(e^{-2\delta T_x}\right) = \int_0^\infty e^{-2\delta T_x}\, {}_tp_x\, \mu_{x+t}\, dt = \int_0^\infty v^{2T_x}\, {}_tp_x\, \mu_{x+t}\, dt.$$

Notice that calculating ${}^2\bar{A}_x$ at the force of interest δ gives the same value as calculating $\bar{A}_x$ at the force of interest 2δ.

Equivalently, calculating ${}^2\bar{A}_x$ at the effective rate of interest i gives the same value as calculating $\bar{A}_x$ at the effective rate of interest $(1+i)^2 - 1$.

The variance of the present value Z of a whole life insurance that pays £1 immediately upon the death of (x) is thus

$$\mathbb{V}ar(Z) = {}^2\bar{A}_x - \left(\bar{A}_x\right)^2.$$

Example 4.5 *A whole life insurance pays £1 immediately upon the death of (65). Given that* $\bar{A}_{65} = 0.52786$ *and* ${}^2\bar{A}_{65} = 0.30855$*, the expected present value is £*0.52786 *and the standard deviation of the present value of the insurance is*

$$\sqrt{{}^2\bar{A}_{65} - \left(\bar{A}_{65}\right)^2} = \sqrt{0.30855 - (0.52786)^2} = \sqrt{0.0299138} = 0.17296.$$

Exercise 4.6 *What is the source of the variance of the present value Z?*

The variance is due to random fluctuations in the time when the actual death occurs against the distribution of the time of death. This is called *idiosyncratic longevity risk* or *idiosyncratic mortality risk* or *random fluctuations risk* (there is not a unique terminology). There is no variance from investment returns as these are assumed to be constant. (Though, in reality, additional variance would arise from volatility in investment returns.)

Remark 4.7 *The variance calculated in the example above assumes that only one policy has been sold. As more policies are sold, the average variance per policy decreases.*

For example, suppose an integer $N > 0$ policies are sold to N lives with future lifetime random variables $T_1, T_2, \ldots, T_N$. Model the present value of policy n by the random variable $Z_n := v^{T_n}$, for $n = 1, 2, \ldots, N$. Assume future lifetimes are independent, which means that $Z_1, Z_2, \ldots, Z_N$ inherit the independence. Further assume future lifetimes each have the same distribution (this is not a critical assumption but will give a simple result), which means Z_1 has the same distribution as $Z_2, Z_3, \ldots, Z_N$. Hence $Z_1, Z_2, \ldots, Z_N$ have the same variance, denoted by $\sigma^2 > 0$.

Then the variance of the average present value is

$$\begin{aligned}
\mathbb{V}ar\left(\frac{Z_1 + Z_2 + \cdots + Z_N}{N}\right) &= \frac{1}{N^2}\mathbb{V}ar(Z_1 + Z_2 + \cdots + Z_N) && \textit{By properties of the variance.} \\
&= \frac{1}{N^2}\left(\mathbb{V}ar(Z_1) + \mathbb{V}ar(Z_2) + \cdots + \mathbb{V}ar(Z_N)\right) && \textit{By independence.} \\
&= \frac{1}{N^2}N\mathbb{V}ar(Z_1) && \textit{As } Z_1, Z_2, \ldots, Z_N \textit{ have the same distribution.} \\
&= \frac{\mathbb{V}ar(Z_1)}{N}.
\end{aligned}$$

As N gets larger, the variance of the average present value will get progressively smaller.

More examples and exercises on whole life insurance payable immediately on death

Example 4.8 *Suppose a whole life insurance pays £100 immediately upon the death of* (x)*. Then it has a random present value*

$$100v^{T_x}.$$

Its expected present value is

$$100\bar{A}_x = \mathbb{E}\left(100v^{T_x}\right).$$

Exercise 4.9 *(a) Show that, for a positive rate of interest,* $0 < \bar{A}_x < 1$.

$\bar{A}_x = \mathbb{E}(v^{T_x}) = \mathbb{E}(e^{-\delta T_x})$ *As the exponential function is always positive, then* $e^{-\delta T_x} > 0$*. Thus its expected value* $\bar{A}_x = \mathbb{E}(v^{T_x}) > 0$*, as taking expectations effectively means multiplying each possible value of* v^{T_x} *by its probability of occurrence – all positive values in this case – and summing them up.*

By the properties of the exponential function and using that $T_x > 0$*,* $e^{-\delta T_x} < 1$*. Hence, as* ${}_tp_x\,\mu_{x+t}$ *is the pdf of* T_x *and must therefore integrate to value 1 over the range of values os* T_x*,*

$$\bar{A}_x = \int_0^\infty e^{-\delta T_x}\,{}_tp_x\,\mu_{x+t}\,dt < \int_0^\infty {}_tp_x\,\mu_{x+t}\,dt1.$$

(b) If we increase the rate of interest used to calculate $\bar{A}_x$*, will its value go up or down?*

The value will go down if the interest rate increases, as less money is needed today to pay the future sum insured.

(c) In general, will $\bar{A}_x$ *be greater or less for a female than for a male, everything else being equal?*

A female is likely to live longer, on average, to a comparable male. This means it is a longer time, on average, until the sum insured is paid. Thus the expected present value $\bar{A}_x$ *should be less for a female.*

(d) Explain why $\mathbb{E}\left(v^{T_x}\right) \neq v^{\mathbb{E}(T_x)}$*, by using Jensen's inequality.*

Jensen's inequality states that, if X *is a random variable and* $g : \mathbb{R} \to \mathbb{R}$ *is a convex function – where* g *is convex if* $g''(t) \geq 0$ *for all* $t \in \mathbb{R}$ *– then* $\mathbb{E}(g(X)) \geq g\left(\mathbb{E}(X)\right)$*. If* g *is strictly convex, i.e.* $g''(t) > 0$*, then Jensen's inequality is strict, i.e.* $\mathbb{E}(g(X)) > g\left(\mathbb{E}(X)\right)$*.*

Consider $g(t) := v^t = e^{-\delta t}$*. Then* g *is a strictly convex function on* $[0, \infty)$ *since for all* $t \in [0, \infty)$*,*

$$g'(t) = -\delta e^{-\delta t} = -\delta g(t) \quad \Rightarrow \quad g''(t) = \delta^2 \left(g(t)\right)^2 > 0.$$

Applying Jensen's inequality,

$$\mathbb{E}\left(v^{T_x}\right) > v^{\mathbb{E}(T_x)}.$$

Therefore these two quantities are not equal.

Exercise 4.10 *A whole life insurance has a sum insured of £S payable immediately on the death of (x), for a constant $S > 0$.*

Write down an expression for the present value of the sum insured. Hence derive an expression for the variance of its present value in terms of $\bar{A}_x$ and ${}^2\bar{A}_x$.

The present value of the sum insured is Sv^{T_x}. Its EPV is $\mathbb{E}(Sv^{T_x}) = S \cdot \bar{A}_x$ and its variance is

$$\mathbb{V}ar(Sv^{T_x}) = S^2 \cdot \mathbb{V}ar(v^{T_x}) = S^2 \cdot \left({}^2\bar{A}_x - \left(\bar{A}_x\right)^2\right).$$

Exercise 4.11 *You are given the following values:*

- $\bar{A}_{40} = 0.23514$ *at an effective rate of interest* $i = 4\%$ *per year, and*
- $\bar{A}_{40} = 0.07065$ *at an effective rate of interest* $i = 8.16\%$ *per year.*

In both cases, the survival model is AM92 Ultimate. Note that $1.0816 = 1.04^2$.

Calculate the expectation and standard deviation of the present value of a payment of £$50,000$ *immediately on the death of* (40)*, whenever that occurs in the future. Assume that the interest rate for calculation is* $i = 4\%$ *per year effective and the survival model is AM92 Ultimate.*

The present value of the payment is £$Z := 50,000v^{T_{40}}$. The expected present value is

$$\mathbb{E}(Z) = 50,000\mathbb{E}\left(v^{T_{40}}\right) = 50,000\ \bar{A}_{40} = 50,000 \times 0.23514 = £11,757.$$

The variance of the present value is

$$\mathbb{V}ar(Z) = (50,000)^2 \times \left(\mathbb{E}\left(v^{2T_{40}}\right) - \left(\mathbb{E}\left(v^{T_{40}}\right)\right)^2\right) = (50,000)^2 \times \left({}^2\bar{A}_{40} - \left(\bar{A}_{40}\right)^2\right),$$

calculated at $i = 4\%$ per annum. Notice that $\bar{A}_{40}$ calculated at $i = 8.16\%$ per year is the same as ${}^2\bar{A}_{40}$ calculated at $i = 4\%$ per year. Thus

$$\mathbb{V}ar(Z) = (50,000)^2 \times \left(0.07065 - (0.23514)^2\right) = £(6,196.61)^2,$$

and the standard deviation is £$6,197$.

4.2.2 Whole life insurance payable at the end of year of death

In practice, an insurance company cannot pay out benefits immediately on death. The insurance company has to be contacted and informed of the claim, they need to verify the death and then pay the monies. All of this takes time. There are also variations of the contract where payment is made at a specific time.

Here we consider a policy which only pays the death benefit at the end of the year in which the policyholder dies.

Exercise 4.12 *If a policy pays the death benefit at the end of the year of death of (x), is it paid out at time K_x or $K_x + 1$?*

It pays out at time $K_x + 1$.

As an example, suppose in a future state of the world that (x) dies on 1 April 2025, corresponding to $T_x = 23.25$ years since the policy was taken out. The death benefit would be paid on 31 December 2025, the end of the year of death, i.e. 24 years after the policy was in-force.

As $T_x = 23.25$ in that state, then $K_x = 23$ (i.e. 1 January 2024) and the payment date corresponds to time $K_x + 1 = 24$ (i.e. 31 December 2024 or 1 January 2025; we consider the end of 31 December 2024 the same as the start of 1 January 2025).

Definition 4.13 *A whole life insurance that pays a sum insured of £1 at the end of year of death of (x) has a random present value*

$$Z := v^{K_x+1}.$$

Its expected present value is represented in international actuarial notation as

$$A_x := \mathbb{E}\left(v^{K_x+1}\right) = \sum_{k=0}^{\infty} v^{k+1}\, {}_{k|}q_x = \sum_{k=0}^{\infty} v^{k+1}\, {}_kp_x\, q_{x+k}.$$

Exercise 4.14 *Show that if $x \leq \omega - 1$ that*

$$A_x = \sum_{k=0}^{\omega-x-1} v^{k+1}\, {}_{k|}q_x.$$

First

$$A_x = \sum_{k=0}^{\infty} v^{k+1}\, {}_{k|}q_x = \sum_{k=0}^{\omega-x-1} v^{k+1}\, {}_{k|}q_x + \sum_{k=\omega-x}^{\infty} v^{k+1}\, {}_{k|}q_x.$$

As ${}_{\omega-x}p_x = {}_{\omega-x+1}p_x = \cdots = 0$, then

$$ {}_{(\omega-x)|}q_x = {}_{\omega-x}p_x q_{x+k} = 0, \quad {}_{(\omega-x+1)|}q_x = {}_{\omega-x+1}p_x q_{x+k} = 0, \quad \ldots, $$

Hence the sum starting from $k = \omega - x$ has value zero and so $A_x = \sum_{k=0}^{\omega-x-1} v^{k+1}\, {}_{k|}q_x$.

Exercise 4.15 *(a) For a whole life insurance of £1 payable at the end of the year of death, show that the variance of the present value of the death benefit is:*

$$ {}^2A_x - (A_x)^2 , $$

where ${}^2A_x$ is the expected present value of a whole life insurance calculated at a rate of interest of $(1+i)^2 - 1$ per year.

(b) If the benefit is a constant £$S > 0$, derive a similar expression for the variance.

(a) For a sum insured of £1,

$$ \mathbb{V}ar\left(v^{K_x+1}\right) = \mathbb{E}\left(v^{2K_x+1}\right) - \mathbb{E}\left(v^{K_x+1}\right)^2 = \mathbb{E}\left(\left(v^2\right)^{K_x+1}\right) - \mathbb{E}\left(v^{K_x+1}\right)^2 = {}^2A_x - (A_x)^2 , $$

where ${}^2A_x$ is the expected present value of a whole life assurance calculated at a rate of interest of $(1+i)^2 - 1$ per year.

(b) For a sum insured of £$S > 0$,

$$ \mathbb{V}ar\left(S \cdot v^{K_x+1}\right) = S^2 \mathbb{V}ar\left(v^{K_x+1}\right) = S^2\left({}^2A_x - (A_x)^2\right). $$

Recursive calculation of A_x

A recursive method to calculate A_x is useful in the construction of tables or in spreadsheet calculations.

Exercise 4.16 *Show the recursion formula*

$$\boxed{A_x = v\,q_x + v\,p_x\,A_{x+1},} \qquad \text{for } x = 0, 1, \ldots, \omega - 1.$$

Hint: Begin with the summation formula for A_x and aim to extract A_{x+1} from the sum.

$$\begin{aligned} A_x = \sum_{k=0}^{\infty} v^{k+1} = \sum_{k=0}^{\infty} v^{k+1}\,{}_kp_x\,q_{x+k} &= v\,q_x + v^2\,p_x\,q_{x+1} + v^3\,{}_2p_x\,q_{x+2} + \cdots \\ &= v\,q_x + v\,p_x\left(vq_{x+1} + v^2\,p_{x+1}\,q_{x+2} + \cdots\right) \\ &= v\,q_x + v\,p_x\,A_{x+1}. \end{aligned}$$

Exercise 4.17 *Suppose you are given a life table and you want to calculate $\{A_x; x = 0, 1, 2, \ldots, \omega\}$.*

How could you do this recursively? Would you start with the value of A_0 or A_ω?

Start at the limiting age ω. As all lives are assumed to be dead at age ω, i.e. $q_{\omega-1} = 1$, then $A_\omega = 0$. Then use the recursion formula to work backwards through the ages, i.e. use $A_x = vq_x + v\,p_x\,A_{x+1}$.

To show some of the calculations that would be performed:

$$\begin{aligned} A_{\omega-1} &= v\,q_{\omega-1} + v\,p_{\omega-1}\,A_\omega = v \times 1 + v \times 0 \times 0 = v. \\ A_{\omega-2} &= v\,q_{\omega-2} + v\,p_{\omega-2}\,A_{\omega-1} = v\,q_{\omega-2} + v^2\,p_{\omega-2}. \\ \vdots &= \vdots \end{aligned}$$

Formulae and Tables for Actuarial Examinations

In the Formulae and Tables for Actuarial Examinations, you will find the formula for the EPV and variance of the whole life insurances paid immediately upon death and at the end of year of death.

4.3 Term insurance

A term insurance pays a benefit (the sum insured) upon death too. However, the benefit is only paid if death occurs before a specified date. In other words, the payment is *contingent* upon death occurring before the specified date.

Term life insurance may be purchased by individuals seeking to protect their dependants financially in the event of their death. It can be used to cover debts, mortgages, or provide income replacement for a specific period.

The term of the policy could be 1 year, 5 years, 25 years,... whatever is agreed between the policyholder and the insurance company. We represent the term by a constant $n > 0$ years.

Notation 4.18 *For an event $A \in \mathcal{F}$, the random indicator function $\mathbb{1}_{[T_x \in A]}$ takes value one for all future states of the world $\omega \in \Omega$ in which $T_x(\omega) \in A$, and value zero if $T_x(\omega) \in \Omega \setminus A$, i.e. for $\omega \in \Omega$,*

$$\mathbb{1}_{[T_x \in A]}(\omega) = \begin{cases} 1 & \text{if } T_x(\omega) \in A, \\ 0 & \text{otherwise.} \end{cases}$$

We normally write this random variable omitting the ω, i.e.

$$\mathbb{1}_{[T_x \in A]} = \begin{cases} 1 & \text{if } T_x \in A, \\ 0 & \text{otherwise.} \end{cases}$$

Note that $\mathbb{1}_{[T_x \in A]}$ is a random variable, being a function of the random variable T_x. The expectation of the random indicator function $\mathbb{1}_{[T_x \in A]}$ is

$$\boxed{\mathbb{E}\left(\mathbb{1}_{[T_x \in A]}\right) = \mathbb{P}[T_x \in A],}$$

which follows from

$$\mathbb{E}\left(\mathbb{1}_{[T_x \in A]}\right) = 1 \times \mathbb{P}[T_x \in A] + 0 \times \mathbb{P}[T_x \notin A] = \mathbb{P}[T_x \in A].$$

4.3.1 n-year term insurance payable immediately on death

Definition 4.19 *Consider an n-year term insurance that pays a sum insured of £1 at the end of year of death of (x), if the death occurs within n years of the policy starting. The benefit has a random present value*

$$Z := v^{T_x}\, \mathbb{1}_{[T_x \le n]} = \begin{cases} v^{T_x} & \text{if } T_x \le n, \\ 0 & \text{otherwise.} \end{cases}$$

Its expected present value is represented in international actuarial notation as

$$\bar{A}^{1}_{x:\overline{n}|} := \mathbb{E}\left(v^{T_x}\, \mathbb{1}_{[T_x \le n]}\right) = \int_0^\infty v^t\, \mathbb{1}_{[t \le n]}\, {}_tp_x \mu_{x+t} dt = \int_0^n e^{-\delta t}\, {}_tp_x \mu_{x+t} dt.$$

The formula for the variance follows the same pattern as before:

$$\mathbb{V}ar\,(Z) = {}^{2}\bar{A}^{1}_{x:\overline{n}|} - \left(\bar{A}^{1}_{x:\overline{n}|}\right)^2,$$

where ${}^{2}\bar{A}^{1}_{x:\overline{n}|}$ *calculated at the force of interest* δ *has the same value as* $\bar{A}^{1}_{x:\overline{n}|}$ *calculated at the force of interest* 2δ*.*

Notation 4.20 *In the symbol* $\bar{A}^{1}_{x:\overline{n}|}$*,*

- *The bent line around the* n *indicates that it is a length of time (in years). In contrast, ages have no other decoration.*
- *There are two implied events that can occur:* (x) *dies and* n *years elapse.*
- *There is the number '1' above the age* x*. It indicates that payment occurs only if the event that* (x) *dies happens first. There is no payment if the* n *years have elapsed first, before* (x) *dies.*

Exercise 4.21 *Suppose* (x) *dies within* n *years of the policy start date. State the possible number of integer years lived by* (x)*, in this case (i.e. state the possible values of* K_x*)?*

$K_x = 0, 1, 2, \ldots, n-1.$

4.3.2 n-year term insurance payable at the end of year of death

Definition 4.22 *Consider an* n*-year term insurance that pays a sum insured of £1 at the end of year of death of* (x)*, if the death occurs within* n *years of the policy starting. The benefit has a random present value*

$$Z := v^{K_x+1}\,\mathbb{1}_{[K_x \le n-1]} = \begin{cases} v^{K_x+1} & \text{if } K_x \le n-1, \\ 0 & \text{otherwise.} \end{cases}$$

Its expected present value is represented in international actuarial notation as

$$A^{1}_{x:\overline{n}|} := \mathbb{E}\left(v^{K_x+1}\,\mathbb{1}_{[K_x \le n-1]}\right) = \sum_{k=0}^{\infty} v^{k+1}\,\mathbb{1}_{[k \le n-1]}\ \cdot {}_{k|}q_x = \sum_{k=0}^{n-1} v^{k+1}\,{}_{k}p_x\,q_{x+k}.$$

The formula for the variance follows the same pattern as before:

$$\mathbb{V}ar\,(Z) = {}^{2}A^{1}_{x:\overline{n}|} - \left(A^{1}_{x:\overline{n}|}\right)^2,$$

where ${}^{2}A^{1}_{x:\overline{n}|}$ *calculated at the interest rate* i *has the same value as* $A^{1}_{x:\overline{n}|}$ *calculated at the interest rate* $(1+i)^2 - 1$*.*

4.4 n-year pure endowment

A pure endowment pays a benefit only if the insured life survives to a specified date. For example, a 10-year pure endowment with sum insured £2,000, issued to John age 50 years, will pay £2,000 in 10 years' time to John only if he is alive then. It pays nothing if John dies between age 50 and 60.

Pure endowment are not sold as stand-alone policies. Instead, they are sold with term insurance policies. This creates another contract called an endowment insurance or endowment assurance, which is studied in the next section.

Definition 4.23 *Consider an n-year pure endowment that pays a sum insured of £1 in exactly n years' time if (x) is alive at age $x+n$. The benefit has a random present value*

$$Z := v^{K_x+1}\,\mathbb{1}_{[T_x>n]} = \begin{cases} 0 & \text{if } T_x \le n, \\ v^n & \text{if } T_x > n. \end{cases}$$

Its expected present value is represented in international actuarial notation as

$$A_{x:\overline{n}|}^{\,1} := \mathbb{E}\left(v^n\,\mathbb{1}_{[T_x>n]}\right) = v^n\,\mathbb{E}\left(\mathbb{1}_{[T_x>n]}\right) = v^n\,\mathbb{P}[T_x > n] = v^n\,{}_np_x.$$

Instead of using $A_{x:\overline{n}|}^{\,1}$, we often use $v^n\,{}_np_x$.

Notation 4.24 *In the symbol $A_{x:\overline{n}|}^{\,1}$,*

- *Again, there are two implied events that can occur: '(x) dies' and 'n years elapse'.*
- *There is the number '1' above the term n. It indicates that payment occurs only if the event that 'n years elapse' happens first. There is no payment if (x) dies first, before n years have elapsed.*
- *Additionally, the payment in the pure endowment is always immediate.*

Exercise 4.25 *Let Z be the present value random variable of a n-year pure endowment that has a sum insured of £1. Show that*

$$\mathbb{V}ar(Z) = v^{2n}\,\left({}_np_x - ({}_np_x)^2\right)$$

and hence

$$\mathbb{V}ar(Z) = v^{2n}\,{}_np_x\,{}_nq_x.$$

We know $\mathbb{V}ar(Z) = \mathbb{E}(Z^2) - (\mathbb{E}Z)^2$ and that $\mathbb{E}(Z) = v^n\,{}_np_x$. As

$$\left(\mathbb{1}_{[T_x>n]}\right)^2 = \mathbb{1}_{[T_x>n]} \cdot \mathbb{1}_{[T_x>n]} = \mathbb{1}_{[T_x>n]},$$

then

$$\mathbb{E}(Z^2) = \mathbb{E}\left(v^{2n}\ \left(\mathbb{1}_{[T_x>n]}\right)^2\right) = v^{2n}\,\mathbb{E}\left(\mathbb{1}_{[T_x>n]}\right) = v^{2n}\,\mathbb{P}[T_x > n] = v^{2n}\ {}_np_x.$$

Finally,

$$\mathbb{V}ar(Z) = \mathbb{E}(Z^2) - (\mathbb{E}Z)^2 = v^{2n}\ {}_np_x - \left(v^n\ {}_np_x\right)^2 = v^{2n}\ \left({}_np_x - ({}_np_x)^2\right).$$

Since ${}_np_x = 1 - {}_nq_x$,

$$\mathbb{V}ar(Z) = v^{2n}\ {}_np_x\ (1 - {}_np_x) = v^{2n}\ {}_np_x\ {}_nq_x.$$

Exercise 4.26 *Set out, in tabular form, the approximations of EPVs of insurance benefits under UDD and claims acceleration. Then sit down and memorise them.*

4.5 Endowment insurance

An endowment insurance policy is the combination of an n-year term insurance policy and an n-year pure endowment policy. It is payable either on:

- Death within n years, or
- Survival to n years,

whichever occurs first. As there is always a payment made in this policy, you will sometimes see it called an *endowment assurance*.

Endowment insurance policies can be used for mortgage-holders. The endowment insurance is used to pay off the capital borrowed when the mortgage matures. Meanwhile, the mortgage-holder pays the interest accumulating on the capital sum as well as the premiums due on the endowment insurance. If the mortgage-holder dies during the term then the endowment insurance will pay off the capital sum too.

There are two types of endowment insurance considered next,depending on when the payment is made when the insured life (x) dies first.

4.5.1 n-year endowment insurance payable immediately on death

Definition 4.27 *Consider an n-year endowment insurance that pays a sum insured of £1 immediately upon the death of (x), if the death occurs within n years of the policy starting, or after exactly n years have elapsed, if (x) is still alive then. The benefit has a random present value*

$$Z := v^{\min\{T_x,n\}} = \begin{cases} v^{T_x} & \text{if } T_x \leq n, \\ v^n & \text{otherwise.} \end{cases}$$

Its expected present value is represented in international actuarial notation as

$$\begin{aligned} \bar{A}_{x:\overline{n}|} :=& \mathbb{E}\left(v^{T_x}\,\mathbb{1}_{[T_x\leq n]}\right) + \mathbb{E}\left(v^n\,\mathbb{1}_{[T_x>n]}\right) \\ =& \bar{A}^1_{x:\overline{n}|} + A_{x:\overline{n}|}^{\,1}. \end{aligned}$$

The second moment of Z is

$$\begin{aligned} {}^2\bar{A}_{x:\overline{n}|} := \mathbb{E}\left(Z^2\right) = \mathbb{E}\left(v^{\min\{2T_x,2n\}}\right) &= \begin{cases} v^{2T_x} & \text{if } T_x \leq n, \\ v^{2n} & \text{otherwise} \end{cases} \\ &= {}^2\bar{A}^1_{x:\overline{n}|} + v^{2n}\,{}_np_x. \end{aligned}$$

The formula for the variance follows the same pattern as before:

$$\mathbb{V}ar\left(Z\right) = {}^2\bar{A}_{x:\overline{n}|} - \left(\bar{A}_{x:\overline{n}|}\right)^2,$$

where ${}^2\bar{A}_{x:\overline{n}|}$ calculated at the force of interest δ has the same value as $\bar{A}_{x:\overline{n}|}$ calculated at the force of interest 2δ.

Notation 4.28 *In the symbol* $\bar{A}_{x:\overline{n}|}$,

- *Again, there are two implied events that can occur:* (x) *dies and* n *years elapse.*
- *There is no number above the age* x *and term* n*. It indicates that the order in which the events occur does not matter. However, payment only occurs once.*
- *The bar above the* A *indicates that, if* (x) *dies first, the payment occurs immediately.*

4.5.2 n-year endowment insurance payable at the end of year of death

Definition 4.29 *Consider an* n*-year endowment insurance that pays a sum insured of £1 either at the end of year of death of* (x)*, if the death occurs within* n *years of the policy starting, or after exactly* n *years have elapsed, if* (x) *is still alive then. The benefit has a random present value*

$$Z := v^{\min\{K_x+1,n\}} = \begin{cases} v^{T_x} & \text{if } K_x \leq n-1, \\ v^n & \text{otherwise.} \end{cases}$$

Its expected present value is represented in international actuarial notation as

$$\begin{aligned} A_{x:\overline{n}|} :=& \mathbb{E}\left(v^{K_x+1}\,\mathbb{1}_{[K_x\leq n-1]}\right) + \mathbb{E}\left(v^n\,\mathbb{1}_{[K_x\geq n]}\right) \\ =& A^1_{x:\overline{n}|} + A_{x:\overline{n}|}. \end{aligned}$$

The second moment of Z *is*

$${}^2A_{x:\overline{n}|} := \mathbb{E}\left(Z^2\right) = \mathbb{E}\left(v^{\min\{2(K_x+1),2n\}}\right) = \begin{cases} v^{2(K_x+1)} & \text{if } K_x \leq n-1, \\ v^{2n} & \text{otherwise} \end{cases}$$
$$= {}^2A^1_{x:\overline{n}|} + v^{2n}\,{}_np_x.$$

The formula for the variance follows the same pattern as before:

$$\mathbb{V}ar\left(Z\right) = {}^2A_{x:\overline{n}|} - \left(A_{x:\overline{n}|}\right)^2,$$

where ${}^2A_{x:\overline{n}|}$ *calculated at the effective rate of interest* i *has the same value as* $A_{x:\overline{n}|}$ *calculated at the effective rate of interest* $(1+i)^2 - 1$*.*

Notation 4.30 *The absence of the bar above the* A *indicates that, if* (x) *dies first, the payment occurs at the end of year of* (x)*'s death.*

4.5.3 Exercises on endowment insurances

Exercise 4.31 *Suppose the effective rate of interest $i > 0$. Explain in words why*

$$A_{x:\overline{n}|} \leq \bar{A}_{x:\overline{n}|}$$

and then prove that it holds mathematically.

If $A_{x:\overline{n}|} \leq \bar{A}_{x:\overline{n}|}$ then the expected present value of the sum insured is paid at the end of year of death is less than when the payment is immediate. It is less because the potential payments of £1 are further into the future; paid at the end of the year rather than immediately.

To show this mathematically, let the present value random variable of the endowment insurance paid:

- Immediately upon death be $\bar{Z} := v^{\min\{T_x, n\}}$.
- At the end of year of death be $Z := v^{\min\{K_x+1, n\}}$.

Since K_x is the integer part of T_x then

$$T_x < K_x + 1, \text{ a.s.} \qquad \Rightarrow \qquad \min\{T_x, n\} \leq \min\{K_x + 1, n\}, \text{ a.s.}$$

Since $1 + i > 1$ then $v = (1 + i)^{-1} < 1$. This means that

$$v^{\min\{T_x, n\}} \geq v^{\min\{K_x+1, n\}}, \text{ a.s.} \qquad \Rightarrow \qquad \bar{Z} \geq Z, \text{ a.s.}$$

Taking expectations on both sides, we get $\mathbb{E}\left(\bar{Z}\right) \geq \mathbb{E}(Z)$. Hence $\bar{A}_{x:\overline{n}|} \geq A_{x:\overline{n}|}$.

Exercise 4.32 *Suppose the effective rate of interest $i > 0$. Explain in words why, for $n, m > 0$,*

$$A_{x:\overline{n}|} \geq A_{x:\overline{n+m}|}.$$

Note than in an endowment insurance, the sum insured is always paid. The question is when is it paid.

An endowment insurance with the shorter term (the n-year term) will have some potential payments paid earlier. These earlier payments correspond to events where (x) survives for at least n years and the sum insured is paid at time n.

In contrast, under the $(n + m)$-year endowment insurance, the sum insured would be paid after time n, as long as (x) survives for at least n years.

The sooner the payment, the higher the present value today. Hence the n-year endowment insurance has the higher expected present value.

To show this mathematically, let the present value random variable of the endowment insurance with a term of:

- n years be $Z^{(n)} := v^{\min\{K_x+1, n\}}$.

- $(n+m)$ years be $Z^{(n+m)} := v^{\min\{K_x+1,n+m\}}$.

Since $m > 0$ then

$$n < n+m \qquad \Rightarrow \qquad \min\{K_x+1, n\} \leq \min\{K_x+1, n+m\}, \text{ a.s.}$$

Since $1+i > 1$ then $v = (1+i)^{-1} < 1$. This means that

$$v^{\min\{K_x+1,n\}} \geq v^{\min\{K_x+1,n+m\}}, \text{ a.s.} \qquad \Rightarrow \qquad Z^{(n)} \geq Z^{(n+m)}, \text{ a.s.}$$

Taking expectations on both sides, we get $\mathbb{E}(Z^{(n)}) \geq \mathbb{E}\left(Z^{(n+m)}\right)$. Hence $A_{x:\overline{n}|} \geq A_{x:\overline{n+m}|}$.

Exercise 4.33 *Show that*

$$A_{x:\overline{n}|} = \sum_{k=0}^{n-2} v^{k+1}\, {}_kp_x q_{x+k} + v^n \cdot {}_{n-1}p_x.$$

As $A_{x:\overline{n}|} = A^1_{x:\overline{n}|} + A_{x:\overline{n}|}^{\,1}$ then substituting for the formula for $A^1_{x:\overline{n}|}$ and $A_{x:\overline{n}|}^{\,1}$ gives

$$\begin{aligned} A_{x:\overline{n}|} &= \sum_{k=0}^{n-1} v^{k+1}\, {}_kp_x\, q_{x+k} + v^n\, {}_np_x \\ &= \sum_{k=0}^{n-2} v^{k+1}\, {}_kp_x\, q_{x+k} + v^n\, {}_{n-1}p_x\, q_{x+n-1} + v^n\, {}_np_x. \end{aligned}$$

Using the multiplicative property of survival probabilities, ${}_np_x = {}_{n-1}p_x \cdot p_{x+n-1}$, and so

$$A_{x:\overline{n}|} = \sum_{k=0}^{n-2} v^{k+1}\, {}_kp_x\, q_{x+k} + v^n\, {}_{n-1}p_x \underbrace{(q_{x+n-1} + p_{x+n-1})}_{=1} = \sum_{k=0}^{n-2} v^{k+1}\, {}_kp_x\, q_{x+k} + v^n \cdot {}_{n-1}p_x.$$

4.6 Deferred insurance benefits

Deferred insurance means that the insurance coverage does not begin until the end of the deferred period. Our main interest in deferred insurances is as a 'building block' to do calculations.

Example 4.34 (Whole life insurance deferred for m years.) *Consider a whole life insurance that is deferred for an integer $m \geq 0$ years and pays a sum insured of £1 at the end of year of death. It only pays the benefit if (x) dies past age $x+m$. It has a present value random variable*

$$Z = \begin{cases} v^{K_x+1} & \text{if } K_x \geq m, \\ 0 & \text{if } K_x \leq m-1. \end{cases}$$

The international actuarial notation for the expected value of Z is

$$ {}_{m|}A_x := \mathbb{E}(Z) = \mathbb{E}\left(v^{K_x+1}\,\mathbb{1}_{[K_x \geq m]}\right) = \sum_{k=m}^{\infty} v^{k+1}\,{}_kp_x\,q_{x+k}. $$

Exercise 4.35 (Whole life insurance deferred for m years.) *Show that*

$$\boxed{ {}_{m|}A_x = v^m\,{}_mp_x\,A_{x+m}. }$$

Explain the result above in words.

As

$$ {}_{m|}A_x = \sum_{k=m}^{\infty} v^{k+1}\,{}_kp_x\,q_{x+k} = v^m\,{}_mp_x \sum_{k=m}^{\infty} v^{k+1-m}\,\frac{{}_kp_x}{{}_mp_x}\,q_{x+k}, $$

change the index of summation to $r = k-m$ to get

$$ {}_{m|}A_x = v^m\,{}_mp_x \sum_{r=0}^{\infty} v^{r+1}\,\underbrace{\frac{{}_{r+m}p_x}{{}_mp_x}}_{={}_rp_{x+m}}\,q_{x+r+m} = v^m\,{}_mp_x \sum_{r=0}^{\infty} v^{r+1}\,{}_rp_{x+m}\,q_{x+r+m} = v^m\,{}_mp_x\,A_{x+m}. $$

A whole life insurance issued to (x) and deferred for m years will not pay out unless (x) dies after age $x+m$. The expected value of the death benefit at age $x+m$ is A_{x+m}. Discounting the value to age x, allowing for survival and interest, gives an expected value at age x of $v^m\,{}_mp_x\,A_{x+m}$. This is equal to ${}_{m|}A_x$, the notation for the EPV of a whole life insurance issued to (x) that is deferred for m years, with the same sum insured of £1.

Other deferred insurances can be defined similarly.

Example 4.36 (n-year term insurance deferred for m years.) *Suppose a benefit of £1 is payable immediately upon the death of (x) only if (x) dies between ages $x+m$ and $x+m+n$, for real values $n, m > 0$. This is a n-year term insurance that is deferred for m years.*

Let the present value random variable of this benefit be

$$Z = \begin{cases} v^{T_x} & \text{if } T_x \in [m, m+n), \\ 0 & \text{otherwise.} \end{cases}$$

The international actuarial notation for the expected value of Z is

$${}_{m|}\bar{A}^{1}_{x:\overline{n}|} := \mathbb{E}(Z) = \mathbb{E}\left(v^{T_x}\,\mathbb{1}_{[T_x\in[m,m+n)]}\right) = \int_m^{m+n} v^t\,{}_tp_x\,\mu_{x+t}\,dt.$$

Exercise 4.37 (n-year term insurance deferred for m years.) *Show that, for $n, m > 0$,*

$$\boxed{{}_{m|}\bar{A}^{1}_{x:\overline{n}|} = v^m\,{}_mp_x\,\bar{A}^{\,1}_{x+m:\overline{n}|}.}$$

Explain the result above in words.

As

$${}_{m|}\bar{A}^{1}_{x:\overline{n}|} = \sum_{k=m}^{m+n-1} v^{k+1}\,{}_kp_x\,q_{x+k} = v^m\,{}_mp_x \sum_{k=m}^{m+n-1} v^{k+1-m}\,\frac{{}_kp_x}{{}_mp_x}\,q_{x+k},$$

change the index of summation to $r = k - m$ to get

$${}_{m|}A_x = v^m\,{}_mp_x \sum_{r=0}^{n-1} v^{r+1}\,\underbrace{\frac{{}_{r+m}p_x}{{}_mp_x}}_{={}_rp_{x+m}}\,q_{x+r+m} = v^m\,{}_mp_x \sum_{r=0}^{n-1} v^{r+1}\,{}_rp_{x+m}\,q_{x+r+m} = v^m\,{}_mp_x\,\bar{A}^{\,1}_{x+m:\overline{n}|}.$$

An n-year term insurance issued to (x) and deferred for m years will not pay out unless (x) dies between ages $x+m$ and $x+m+n$. The expected value at age $x+m$ of an n-year term insurance that also starts at age $x+m$ is $\bar{A}^{\,1}_{x+m:\overline{n}|}$. Discounting the value to age x, allowing for survival and interest, gives an expected value at age x of $v^m\,{}_mp_x\,\bar{A}^{\,1}_{x+m:\overline{n}|}$. This is equal to ${}_{m|}\bar{A}^{1}_{x:\overline{n}|}$, the notation for the EPV of an n-year term insurance issued to (x) that is deferred for m years, with the same sum insured of £1.

Exercise 4.38 *(a) Show that for $m, n \geq 1$,*

$$\boxed{A^{1}_{x:\overline{m+n}|} = A^{1}_{x:\overline{m}|} + {}_{m|}A^{1}_{x:\overline{n}|}.}$$

Using a change of variable of summation, from r to $k := r - m$, to move from the first line to the second line below,

$$\begin{aligned} A^{1}_{x:\overline{m+n}|} = \sum_{r=0}^{m+n-1} v^{r+1}\,{}_rp_x\,q_{x+r} &= \sum_{r=0}^{m-1} v^{r+1}\,{}_rp_x\,q_{x+r} + \sum_{r=m}^{m+n-1} v^{r+1}\,{}_rp_x\,q_{x+r} \\ &= A^{1}_{x:\overline{m}|} + v^m\,{}_mp_x \sum_{k=0}^{n-1} v^{k+1}\,{}_kp_{x+m}\,q_{x+m+k} \\ &= A^{1}_{x:\overline{m}|} + v^m\,{}_mp_x\,A^{\,1}_{x+m:\overline{n}|} \\ &= A^{1}_{x:\overline{m}|} + {}_{m|}A^{1}_{x:\overline{n}|}. \end{aligned}$$

(b) *Show that*

$$\bar{A}_x = \sum_{r=0}^{\infty} {}_{r|}\bar{A}^{1}_{x:\overline{1}|}.$$

Using a change of variable of integration, from t to $s := t - r$, in the first line,

$$\begin{aligned}\bar{A}_x = \int_0^{\infty} v^t \, {}_tp_x \, \mu_{x+t} \, dt = \sum_{r=0}^{\infty} \int_r^{r+1} v^t \, {}_tp_x \, \mu_{x+t} \, dt &= \sum_{r=0}^{\infty} v^r \, {}_rp_x \int_0^1 v^s \, {}_sp_{x+r} \, \mu_{x+r+s} \, ds \\ &= \sum_{r=0}^{\infty} v^r \, {}_rp_x \, \bar{A}^{1}_{x+r:\overline{1}|} \\ &= \sum_{r=0}^{\infty} {}_{r|}\bar{A}^{1}_{x:\overline{1}|}.\end{aligned}$$

Thus a whole life insurance is equivalent to consecutive one-year term insurance policies, each written on the same policyholder and with the same sum insured.

(c) *Derive a formula for $A_{x:\overline{n}|}$ in terms of $\{{}_{r|}A^{1}_{x:\overline{1}|}\}_{r=0,1,\ldots,n-1}$ and ${}_{n-1|}A^{\frac{1}{1}}_{x:\overline{1}|}$.*
Note that for $r \geq 0$, $A^{1}_{x+r:\overline{1}|} = vq_{x+r}$ and $A^{\frac{1}{1}}_{x+r:\overline{1}|} = v\,p_{x+r}$. Then

$$\begin{aligned}A_{x:\overline{n}|} = \sum_{r=0}^{n-1} v^{r+1} \, {}_rp_x \, q_{x+r} + v^n \, {}_np_x &= \sum_{r=0}^{n-1} v^r \, {}_rp_x \, v \, q_{x+r} + v^{n-1} \, {}_{n-1}p_x \, v \, p_{x+n-1} \\ &= \sum_{r=0}^{n-1} v^r \, {}_rp_x \, A^{1}_{x+r:\overline{1}|} + v^{n-1} \, {}_{n-1}p_x \, A^{\frac{1}{1}}_{x+n-1:\overline{1}|} \\ &= \sum_{r=0}^{n-1} {}_{r|}A^{1}_{x:\overline{1}|} + {}_{n-1|}A^{\frac{1}{1}}_{x:\overline{1}|}.\end{aligned}$$

4.7 m-thly insurance benefits

Insurance benefits which are paid immediately on death or at the end of the year of death are the easiest for us to handle mathematically. However, real life insurance contracts are often more complicated. One complication is that benefits are often paid at the end of the month of death for instance. We handle all the different possible payment frequencies by thinking about benefits paid at the end of the mth of a year of death. (For instance, $m = 12$ where benefits are paid at the end of the month of death.)

4.7.1 Curtate future lifetime random variable $K_x^{(m)}$

Let m be a positive integer. The random variable $K_x^{(m)}$ is the future lifetime of (x), expressed in years, rounded to the lower $1/m$th of a year, i.e.

$$K_x^{(m)} := \frac{1}{m} \lfloor mT_x \rfloor,$$

where $\lfloor mT_x \rfloor$ is the integer part of mT_x. Note that $K_x^{(1)} = K_x$, the curtate future lifetime random variable introduced in Chapter 2.

Example 4.39 *If in some future state of the world ω, a life (5) dies at age 5.4 years. Then $T_5(\omega) = 0.4$ years, whereas the curtate future lifetime rounded to the lowest month is*

$$K_5^{(12)}(\omega) = \frac{1}{12} \lfloor 12(0.4) \rfloor = \frac{1}{12} \lfloor 4.8 \rfloor = \frac{1}{12} (4) = \frac{4}{12} \text{ years}$$

and, rounded to the lowest quarter-year,

$$K_5^{(4)}(\omega) = \frac{1}{4} \lfloor 4(0.4) \rfloor = \frac{1}{4} \lfloor 1.6 \rfloor = \frac{1}{4} (1) = \frac{1}{4} \text{ years.}$$

Note that $K_x^{(m)}$ is a discrete random variable, which takes values $0, \frac{1}{m}, \frac{2}{m}, \ldots$. Let $k \in \{0, \frac{1}{m}, \frac{2}{m}, \ldots\}$. Then $K_x^{(m)} = k$ means that (x) dies in the age range $[x+k, x+k+1/m)$.

The probability mass function of $K_x^{(m)}$ is for $k = 0, \frac{1}{m}, \frac{2}{m}, \ldots,$

$$\boxed{\mathbb{P}[K_x^{(m)} = k] = \mathbb{P}\left[k \le T_x < k + \frac{1}{m}\right] = {}_{k|\frac{1}{m}}q_x = {}_kp_x - {}_{k+\frac{1}{m}}p_x.}$$

4.7.2 Whole life insurance payable at the end of month of death

Definition 4.40 *A whole life insurance that pays a sum insured of £1 at the end of the $1/m$-th of a year in which (x) dies has a random present value*

$$v^{K_x^{(m)}+\frac{1}{m}}.$$

Its expected present value is represented in international actuarial notation as

$$A_x^{(m)} := \mathbb{E}\left(v^{K_x^{(m)}+\frac{1}{m}}\right) = \sum_{k=0}^{\infty} v^{\frac{k+1}{m}}\, {}_{\frac{k}{m}|\frac{1}{m}}q_x.$$

Exercise 4.41 *Check that $A_x^{(1)} = A_x$.*

Exercise 4.42 *(a) Explain why, for an interest rate $i > 0$ and a given survival model,*

$$A_x < A_x^{(12)} < \bar{A}_x.$$

Moving from left to right in the inequality, the payment occurs earlier:

- *Under A_x, the payment is at the end of year of death,*
- *Under $A_x^{(12)}$, the payment is at the end of month of death, and*
- *Under $\bar{A}_x$, the payment is immediately upon death, which is the earliest it can occur.*

With a positive interest rate, the discounted death benefits will be largest under $\bar{A}_x$, since they occur the soonest and so there is less time for interest to accumulate. They are smallest under A_x, since they occur the latest.

The expected present values also keep the same ordering, since the probability of each death benefit being paid is the same under the three different insurances. Thus $A_x < A_x^{(12)} < \bar{A}_x$.

(b) Explain what $A^{(4)}_{x:\overline{n}|}$ represents.

$A^{(4)}_{x:\overline{n}|}$ represents the expected present value of an n-year endowment insurance, where the death benefit is paid at the end of the quarter-year of death. The survival benefit (the pure endowment element) is paid upon survival for n years.

(c) Explain what $A^{1\,(12)}_{x:\overline{n}|}$ represents.

$A^{1\,(12)}_{x:\overline{n}|}$ represents the expected present value of an n-year term insurance, where the death benefit is paid at the end of the month of death.

(d) Show that

$$\mathbb{V}ar\left(v^{K_x^{(m)}+\frac{1}{m}}\right) = {}^2A_x^{(m)} - \left(A_x^{(m)}\right)^2,$$

where ${}^2A_x^{(m)}$ *is calculated at the effective rate of interest* $(1+i)^2 - 1$ *per annum.*

Follow the idea in the proof of Exercise 4.15.

4.8 Approximating $\bar{A}_x$ and $A_x^{(m)}$ from A_x

Life tables are specified at integer ages. We can use a life table to calculate the EPV of benefits that are paid at the end of the year of death or on survival to an exact age, such as A_x. What if we want to calculate the EPV of benefits that are paid m-thly or immediately on death, such as $\bar{A}_x$ and $A_x^{(m)}$, and the only data source is a life table?

We study two different approaches to approximating what happens between integer ages, which then allows us to approximate $\bar{A}_x$ and $A_x^{(m)}$ from knowledge of A_x. The first approach is based on a uniform distribution of deaths (UDD) assumption between integer ages. The second follows an argument based on when benefits are paid on average between integer ages.

4.8.1 Approximation using the uniform distribution of deaths (UDD)

Assume that deaths are uniformly distributed between integer ages. This assumption allows us to calculate an exact expression for $\bar{A}_x$ in terms of A_x. So if we calculate A_x from a life table, then we can also calculate $\bar{A}_x$.

Let δ be the annual force of interest, with $e^{-\delta} = (1+i)^{-1} = v$. Consider for integer $x \geq 0$ and integer $n \geq 1$,

$$\bar{A}^1_{x:\overline{n}|} = \int_0^n e^{-\delta t}\, {}_tp_x\, \mu_{x+t}\, dt = \sum_{k=0}^{n-1} \int_k^{k+1} e^{-\delta t}\, {}_tp_x\, \mu_{x+t}\, dt,$$

in which we break up the integral into intervals of length one year so that we can apply the UDD assumption within each interval. Changing the variable of integration from t to $s := t-k$,

$$\bar{A}^1_{x:\overline{n}|} = \sum_{k=0}^{n-1} \int_0^1 e^{-\delta(s+k)}\, {}_{s+k}p_x\, \mu_{x+k+s}\, ds.$$

As ${}_{s+k}p_x = {}_kp_x\, {}_sp_{x+k}$,

$$\bar{A}^1_{x:\overline{n}|} = \sum_{k=0}^{n-1} e^{-\delta k}\, {}_kp_x \int_0^1 e^{-\delta s}\, {}_sp_{x+k}\, \mu_{x+k+s}\, ds.$$

Now apply the UDD assumption to the integrand, which we can do as $x+k$ is an integer and $s \in [0,1)$. From Lemma 3.21, the UDD assumption gives that ${}_sp_{x+k}\, \mu_{x+k+s} = q_{x+k}$. Substituting this result into the integrand, and using that q_{x+k} is independent of the variable of integration s.

$$\bar{A}^1_{x:\overline{n}|} = \sum_{k=0}^{n-1} e^{-\delta k}\, {}_kp_x \int_0^1 e^{-\delta s}\, q_{x+k}\, ds = \sum_{k=0}^{n-1} e^{-\delta k}\, {}_kp_x\, q_{x+k} \int_0^1 e^{-\delta s}\, ds.$$

As

$$\int_0^1 e^{-\delta s}\, ds = \left[\frac{e^{-\delta s}}{-\delta}\right]_{s=0}^{s=1} = \frac{e^{-\delta} - e^0}{-\delta} = \frac{1-e^{-\delta}}{\delta},$$

we get

$$\begin{aligned}\bar{A}^1_{x:\overline{n}|} = \left(\frac{1-e^{-\delta}}{\delta}\right) \sum_{k=0}^{n-1} e^{-\delta k}\, {}_kp_x\, q_{x+k} &= \left(\frac{1-e^{-\delta}}{\delta}\right) \sum_{k=0}^{n-1} e^{-\delta(k+1)}\, {}_kp_x\, q_{x+k} \\ &= \left(\frac{1-e^{-\delta}}{\delta}\right) e^{\delta} \sum_{k=0}^{n-1} e^{-\delta(k+1)}\, {}_kp_x\, q_{x+k} \\ &= \bar{A}^1_{x:\overline{n}|} = \left(\frac{e^{\delta}-1}{\delta}\right) A^1_{x:\overline{n}|},\end{aligned}$$

where the last line follows upon substituting $A^1_{x:\overline{n}|} = \sum_{k=0}^{n-1} e^{-\delta(k+1)}\, {}_kp_x\, q_{x+k}$.

Finally, since $1+i = e^{\delta}$,

$$\boxed{\bar{A}^1_{x:\overline{n}|} = \frac{i}{\delta}\, A^1_{x:\overline{n}|}.}$$

Exercise 4.43 *From the Formulae and Tables for Actuarial Examinations 2025, at an effective interest rate of 6% per annum and on the mortality basis TFNL16,* $A^1_{50:\overline{15}|} = 0.31715$. *Calculate* $\bar{A}^1_{50:\overline{15}|}$ *assuming a uniform distribution of deaths between integer ages.*

Using the result derived above,

$$\bar{A}^1_{50:\overline{15}|} = \frac{i}{\delta}\, A^1_{50:\overline{15}|} = \frac{0.06}{\ln(1.06)}\,(0.31715) = 0.32657.$$

As a check of reasonableness on the numerical answer, note that $\bar{A}^1_{50:\overline{15}|} > A^1_{50:\overline{15}|}$ (an extension of Exercise 4.39(a), using exactly the same reasoning), which is satisfied by $0.32657 > 0.31715$. Also, $0.32657 \approx 0.31715$; the answers are of a similar magnitude since there are only differences in the timing during each year of the payment of the death benefit.

Handy hint 4.44 *In the Formulae and Tables for Actuarial Examinations,* $A^1_{x:\overline{n}|}$ *is tabulated for* $x+n = 65$ *and* $x+n = 60$. *For example, for TFNL16 at 6% p.a. it shows* $A^1_{18:\overline{42}|}, A^1_{19:\overline{41}|}, \ldots, A^1_{59:\overline{1}|}$ *on one page, and* $A^1_{18:\overline{47}|}, A^1_{19:\overline{46}|}, \ldots, A^1_{64:\overline{1}|}$ *on another.*

Many questions based on the Tables assumed that you will look up these values. Make sure you do, rather than attempting to calculate, e.g. $A^1_{50:\overline{15}|}$ *using the values* $\{\ell_x; x = 50, 51, \ldots\}$ *- it will take a lot of time!*

Either by re-working the derivation above for a whole life insurance, or by letting $n \to \infty$ in $\bar{A}^1_{x:\overline{n}|} = \frac{i}{\delta}\, A^1_{x:\overline{n}|}$, we obtain

$$\boxed{\bar{A}_x = \frac{i}{\delta}\, A_x.}$$

Exercise 4.45 *From the Formulae and Tables for Actuarial Examinations 2025, at an effective interest rate of 4% per annum and on the mortality basis TMNL16,* $A_{40} = 0.17311$. *Calculate* $\bar{A}_{40}$ *assuming a uniform distribution of deaths between integer ages.*

Using the result derived above,

$$\bar{A}_{40} = \frac{i}{\delta}\, A_{40} = \frac{0.04}{\ln(1.04)}\,(0.17311) = 0.17655.$$

As a check of reasonableness on the numerical answer, note from Exercise 4.39(a) that $\bar{A}_{40} > A_{40}$, which is satisfied by $0.17655 > 0.17311$. As before, $0.17655 \approx 0.17311$; the answers are close in value since there are only differences in the timing during each year of the payment of the death benefit.

For an n-year endowment insurance, we need to be careful. The pure endowment element is unchanged by the assumption on what happens between integer ages, as it is paid exactly n years after the policy starts.

Assuming UDD between integer ages,

$$\boxed{\bar{A}_{x:\overline{n}|} = \bar{A}^{1}_{x:\overline{n}|} + {}_{n}p_x\, v^n \overset{\text{Apply UDD}}{=} \frac{i}{\delta}\, A^{1}_{x:\overline{n}|} + {}_{n}p_x\, v^n.}$$

Exercise 4.46 *From the Formulae and Tables for Actuarial Examinations 2025, at an effective interest rate of 6% per annum and on the mortality basis TFNL16,* $A^{1}_{50:\overline{15}|} = 0.31715$. *Additionally,* $\ell_{50} = 9903.0603$ *and* $\ell_{65} = 9636.5253$.

Calculate $\bar{A}_{50:\overline{15}|}$ *assuming a uniform distribution of deaths between integer ages.*

Using the result derived above,

$$\begin{aligned}\bar{A}_{50:\overline{15}|} = \frac{i}{\delta}\, A^{1}_{50:\overline{15}|} + {}_{15}p_{50}\, v^{15} &= \frac{0.06}{\ln(1.06)}\,(0.31715) + \frac{9636.5253}{9903.0603}\,(1.06)^{-15} \\ &= 0.326572 + 0.406035 \\ &= 0.73261.\end{aligned}$$

You can show that for a $m$thly-paid whole life insurance, under the UDD assumption,

$$\boxed{A^{(m)}_{x} = \frac{i}{i^{(m)}}\, A_x,}$$

where

$$\left(1 + \frac{i^{(m)}}{m}\right)^{m} = 1 + i.$$

Similarly, under the UDD assumption,

$$\boxed{A^{1(m)}_{x:\overline{n}|} = \frac{i}{i^{(m)}}\, A^{1}_{x:\overline{n}|} \qquad \text{and} \qquad A^{(m)}_{x:\overline{n}|} = \frac{i}{i^{(m)}}\, A^{1}_{x:\overline{n}|} + {}_{n}p_x\, v^n.}$$

4.8.2 Approximation using claims acceleration

The claims acceleration approach approximates all the potential payment times in a year by a single time in that year. Insurances are valued by assuming that a death benefit payment payable in a particular year is made at that single time. However, the pure endowment benefit continues to be paid at time n.

Suppose death benefit payments are made at the end of each $m$thly period, so that potential payment times are $\{0, \frac{1}{m}, \frac{2}{m}, \ldots\}$. Counting from the start of a year, the potential payment times within that year only are $\{\frac{1}{m}, \frac{2}{m}, \ldots, \frac{m}{m}\}$. In that year, calculate the average time of payment as the actual time of payment weighted by $\frac{1}{m}$ to get

$$\frac{1}{m}\cdot\frac{1}{m}+\frac{2}{m}\cdot\frac{1}{m}+\cdots+\frac{m}{m}\cdot\frac{1}{m}=\frac{1}{m^2}\sum_{k=1}^{m}k=\frac{1}{m^2}\,\frac{m(m+1)}{2}=\frac{m+1}{2m}.$$

Assume any death benefit payment that occurs in that year will be paid $\frac{m+1}{2m}$ years from the start of that year. (Notice that $\frac{m+1}{2m}\approx\frac{1}{2}$, so the payments are assumed to occur roughly half-way through the year.)

Value the insurance benefit under this assumption, i.e. the EPV of a whole life insurance payable at the end of the $m$thly period of death is

$$A_x^{(m)}=v^{\frac{m+1}{2m}}\,q_x+v^{1+\frac{m+1}{2m}}\,{}_{1|}q_x+v^{2+\frac{m+1}{2m}}\,{}_{2|}q_x+\cdots=\sum_{k=0}^{\infty}v^{k+\frac{m+1}{2m}}\,{}_{k|}q_x=v^{\frac{m+1}{2m}-1}\sum_{k=0}^{\infty}v^{k+1}\,{}_{k|}q_x.$$

As $v^{\frac{m+1}{2m}-1}=v^{\frac{m+1-2m}{2m}}=v^{-\frac{m-1}{2m}}=(1+i)^{\frac{m-1}{2m}}$ and $A_x=\sum_{k=0}^{\infty}v^{k+1}\,{}_{k|}q_x$, under claims acceleration,

$$\boxed{A_x^{(m)}=(1+i)^{\frac{m-1}{2m}}\,A_x.} \tag{1}$$

The expression for the n-year term insurance follows similarly,

$$\boxed{A_{x:\overline{n}|}^{1\,(m)}=(1+i)^{\frac{m-1}{2m}}\,A_{x:\overline{n}|}^{1}.} \tag{2}$$

For the n-year endowment insurance, the EPV under claims acceleration is

$$\boxed{A_{x:\overline{n}|}^{(m)}=A_{x:\overline{n}|}^{1\,(m)}+{}_np_x\,v^n\overset{\text{Claim acc.}}{=}(1+i)^{\frac{m-1}{2m}}\,A_{x:\overline{n}|}^{1}+{}_np_x\,v^n.}$$

Exercise 4.47 *From the Formulae and Tables for Actuarial Examinations 2025, at an effective interest rate of 6% per annum and on the mortality basis TFNL16,* $A_{50:\overline{15}|}^{1}=0.31715$. *Additionally,* $\ell_{50}=9903.0603$ *and* $\ell_{65}=9636.5253$.

Calculate $A_{50:\overline{15}|}^{(12)}$ *using the claims acceleration approach.*

Using the result derived above,

$$\begin{aligned}A_{50:\overline{15}|}^{(12)}=(1+i)^{\frac{11}{24}}\,A_{50:\overline{15}|}^{1}+{}_{15}p_{50}\,v^{15}&=(1.06)^{\frac{11}{24}}\,(0.31715)+\frac{9636.5253}{9903.0603}\,(1.06)^{-15}\\&=0.325734+0.406035\\&=0.73177.\end{aligned}$$

Now turn to calculating the EPV of insurances where the death benefit is paid immediately upon death. As

$$\lim_{m\to\infty}\left\{(1+i)^{\frac{m-1}{2m}}\,A_x\right\} = A_x\,(1+i)^{\frac{1}{2}}\cdot\lim_{m\to\infty}\left\{(1+i)^{-\frac{1}{2m}}\right\} = A_x\,(1+i)^{\frac{1}{2}}\cdot 1 = A_x\,(1+i)^{\frac{1}{2}},$$

and $\lim_{m\to\infty} A_x^{(m)} = \bar{A}_x$, take the limit as $m\to\infty$ in equation (1) to obtain

$$\boxed{\bar{A}_x = (1+i)^{0.5}\,A_x.}$$

Similarly, as $\lim_{m\to\infty} A^{1\,(m)}_{x:\overline{n}|} = \bar{A}^1_{x:\overline{n}|}$, take the limit as $m\to\infty$ in equation (2) to obtain

$$\boxed{\bar{A}^1_{x:\overline{n}|} = (1+i)^{0.5}\,A^1_{x:\overline{n}|} \qquad \text{and} \qquad \bar{A}_{x:\overline{n}|} = (1+i)^{0.5}\,A^1_{x:\overline{n}|} + {}_np_x\,v^n.}$$

Exercise 4.48 *From the Formulae and Tables for Actuarial Examinations 2025, at an effective interest rate of 4% per annum and on the mortality basis TMNL16,* $A_{40} = 0.17311$. *Calculate* $\bar{A}_{40}$ *using the claims acceleration approach.*

Using the result derived above,

$$\bar{A}_{40} = (1+i)^{0.5}\,A_{40} = (1.04)^{0.5}\,(0.17311) = 0.17654.$$

Compare to the answer using UDD: 0.17655; almost the same value. The same checks apply to your answer, to try to catch calculator or transcription errors.

Exercise 4.49 *From the Formulae and Tables for Actuarial Examinations 2025, at an effective interest rate of 6% per annum and on the mortality basis TFNL16,* $A^1_{50:\overline{15}|} = 0.31715$. *Calculate* $\bar{A}^1_{50:\overline{15}|}$ *using the claims acceleration approach.*

Using the result derived above,

$$\bar{A}^1_{50:\overline{15}|} = (1+i)^{0.5}\,A^1_{50:\overline{15}|} = (1.06)^{0.5}\,(0.31715) = 0.32653.$$

Compare to the answer using UDD: 0.32657.

4.9 Variable insurance benefits

So far, the death benefit and survival benefit amounts have been identical and constant values. Now consider insurances in which these can vary. We assume that the benefits depend only on the future lifetime random variable T_x of the insured life (x), for example, a death benefit which is of amount $£T_x$.

4.9.1 General idea of valuing variable insurance benefits

Mathematically, let the benefit amount be $£h(T_x)$, where $h : [0, \infty) \to [0, \infty)$ is a positive, real-valued function. Suppose the benefit is paid immediately upon death. The present value random variable, Z, of such a benefit is

$$Z := v^{T_x}\, h(T_x).$$

The expected present value of the benefit is

$$\mathbb{E}(Z) := \mathbb{E}\left(v^{T_x}\, h(T_x)\right) = \int_0^\infty v^t\, h(t)\, {}_tp_x\, \mu_{x+t}\, dt$$

and the variance is $\mathbb{V}ar(Z) = \mathbb{E}(Z^2) - \mathbb{E}(Z)^2$, i.e.

$$\mathbb{V}ar(Z) := \mathbb{E}\left(v^{2T_x}\, h^2(T_x)\right) - \mathbb{E}\left(v^{h(T_x)}\right)^2.$$

Example 4.50 *So far, we have seen:*

- *The whole life insurance paying a sum insured of £1 immediately upon death, which means*
$$h(T_x) = 1, \quad \textit{for all } t \geq 0.$$
- *The n-year term insurance paying a sum insured of £1 immediately only if death occurs within n years, which means*
$$h(T_x) = \mathbb{1}_{[T_x \leq n]} = \begin{cases} 1 & \textit{if } T_x \leq n \\ 0 & \textit{otherwise.} \end{cases}$$
- *The n-year endowment insurance paying a sum insured of £1 immediately if death occurs within n years, or £1 immediately upon survival for n years*
$$h(T_x) = \mathbb{1}_{[T_x \leq n]} + v^{n-T_x} \cdot \mathbb{1}_{[T_x > n]} = \begin{cases} 1 & \textit{if } T_x \leq n \\ v^{n-T_x} & \textit{otherwise,} \end{cases}$$
where h is chosen so that
$$h(T_x)v^{T_x} = \begin{cases} 1 \cdot v^{T_x} = v^{T_x} & \textit{if } T_x \leq n \\ v^{n-T_x} \cdot v^{T_x} = v^n & \textit{otherwise.} \end{cases}$$

Example 4.51 *Consider an insurance with term 20 years. If the policyholder dies in the first 8 years, it pays out £5,000 after 8 years. If the policyholder dies in the next 12 years, then it pays out £15,000 after 20 years.*

The present value random variable, £Z, of the insurance is

$$Z := 5,000\, v^8\, \mathbb{1}_{[T_x \le 8]} + 15,000\, v^{20}\, \mathbb{1}_{[8 < T_x \le 20]}.$$

The expected present value of the insurance is

$$\mathbb{E}(Z) = 5,000\, v^8\, \mathbb{P}[T_x \le 8] + 15,000\, v^{20}\, \mathbb{P}[8 < T_x \le 20] = 5,000\, v^8\, {}_8q_x + 15,000\, v^{20}\, {}_{8|12}q_x.$$

Example 4.52 *Consider another insurance with term 20 years. If the policyholder dies in the first 8 years, it pays out £5,000 immediately upon death. If the policyholder dies in the next 12 years, then it pays out £15,000.*

The present value random variable, £Z, of the insurance is

$$Z := 5,000\, \mathbb{1}_{[T_x \le 8]}\, v^{T_x} + 15,000\, \mathbb{1}_{[8 < T_x \le 20]}\, v^{T_x}.$$

To calculate the expected present value of the insurance, we can calculate directly the expectation of Z. Alternatively, consider the insurance as the combination of a 20-year term insurance with sum insured £5,000 and a 12-year term insurance with sum insured £10,000 that is deferred for 8 years, i.e.

$$\mathbb{E}(Z) = 5,000\, \bar{A}^1_{x:\overline{20|}} + 10,000\, {}_{8|}\bar{A}^1_{x:\overline{12|}}.$$

Can you think of another combination of term insurances that gives the same EPV?

Exercise 4.53 *1. Describe in words (how much benefit and when is it paid - both the timing and the event that must occur for it to be payable, i.e. the contingent event) the insurance benefits with the present value random variables given below.*

(a)

$$Z_1 = \begin{cases} 20v^{T_x} & \text{if } T_x \in (0, 15], \\ 10v^{T_x} & \text{if } T_x > 15. \end{cases}$$

It pays:

- *20 units immediately upon (x) dying in the first 15 years, and*
- *10 units immediately upon (x) dying, if (x) dies after at least 15 years have elapsed.*

(b)

$$Z_2 = \begin{cases} 0 & \text{if } T_x \in (0, 5], \\ 10v^{T_x} & \text{if } T_x \in (5, 15], \\ 10v^{15} & \text{if } T_x > 15. \end{cases}$$

It pays:

- *Nothing if (x) dies in the first 5 years, and*
- *10 units immediately upon (x) dying after 5 years have elapsed but before 15 years have elapsed, and*
- *10 units immediately if (x) survives for 15 years.*

2. *Write down in integral form the formula for the expected value for: (i) Z_1, and (ii) Z_2.*

$$\mathbb{E}(Z_1) = 20 \int_0^{15} v^t \, {}_tp_x \, \mu_{x+t} \, dt + 10 \int_{15}^{\infty} v^t \, {}_tp_x \, \mu_{x+t} \, dt.$$

and

$$\mathbb{E}(Z_2) = 10 \int_5^{15} v^t \, {}_tp_x \, \mu_{x+t} \, dt + 10 v^{15} \, {}_{15}p_x.$$

3. *Derive an expression in terms of standard actuarial functions for the expected values of: (i) Z_1, and (ii) Z_2.*

Any of the expressions below are correct.

$$\mathbb{E}(Z_1) = 20 \, \bar{A}^1_{x:\overline{15}|} + 10 \, {}_{15|}\bar{A}_x = 10 \, \bar{A}_x + 10 \, \bar{A}^1_{x:\overline{15}|}.$$

and

$$\mathbb{E}(Z_2) = 10 \, {}_{5|}\bar{A}^1_{x:\overline{10}|} + 10 \, v^{15} \, {}_{15}p_x = 10 \, {}_{5|}\bar{A}^1_{x:\overline{10}|} + 10 \, A_{x:\overline{15}|}^{\;\;1} = 10 \, {}_{5|}\bar{A}^1_{x:\overline{10}|} + 10 \, {}_{15}E_x.$$

4. *Derive an expression in terms of standard actuarial functions for the covariance of Z_1 and Z_2.*

Recall that the covariance of two random variables Z_1 and Z_2 is

$$\mathbb{C}ov(Z_1, Z_2) = \mathbb{E}\left(Z_1 \, Z_2\right) - \mathbb{E}\left(Z_1\right) \mathbb{E}\left(Z_2\right).$$

We've already calculated $\mathbb{E}\left(Z_1\right)$ and $\mathbb{E}\left(Z_2\right)$, so it remains to calculate $\mathbb{E}\left(Z_1 Z_2\right)$. First,

$$Z_1 \, Z_2 = \begin{cases} 20 v^{T_x} \times 0 = 0 & \text{if } T_x \in (0, 5], \\ 20 v^{T_x} \times 10 v^{T_x} = 200 v^{2T_x} & \text{if } T_x \in (5, 15], \\ 10 v^{T_x} \times 10 v^{15} = 100 v^{T_x + 15} & \text{if } T_x > 15. \end{cases}$$

Then

$$\begin{aligned} \mathbb{E}\left(Z_1 Z_2\right) =& 200 \int_5^{15} v^{2t} \, {}_tp_x \, \mu_{x+t} \, dt + 100 \int_{15}^{\infty} v^{t+15} \, {}_tp_x \, \mu_{x+t} \, dt \\ =& 200 \int_5^{15} v^{2t} \, {}_tp_x \, \mu_{x+t} \, dt + 100 \, v^{15} \int_{15}^{\infty} v^t \, {}_tp_x \, \mu_{x+t} \, dt \\ =& 200 \, {}^{2}_{5|}\bar{A}^1_{x:\overline{10}|} + 100 \, v^{15} \, {}_{15|}\bar{A}_x. \end{aligned}$$

4.9.2 International actuarial notation for particular variable insurance benefits

There are some particular variable insurance benefits for which international actuarial notation exists. You have to know the notation, what it represents as well as the associated formulae.

Benefits increasing annually and paid at the end of year of death

Example 4.54 (Whole life insurance paid at the end of year) *Suppose a whole life insurance pays a sum insured of £(K_x+1) at the end of year of death. The present value random variable, Z, of the insurance is*

$$Z := v^{K_x+1}\,(K_x+1).$$

The expected present value of the benefit is

$$\boxed{(IA)_x := \sum_{k=0}^{\infty} (k+1)\; v^{k+1}\;{}_{k|}q_x.}$$

Handy hint 4.55 *Make sure you know that notation, such as $(IA)_x$, and what it represents. By the end of the course, aim to write down the sum for $(IA)_x$ without learning it by rote. So take some time now to understand the composition of the sum.*

For example, in $(IA)_x = \sum_{k=0}^{\infty} (k+1)\; v^{k+1}\;{}_{k|}q_x$,

- *The term $(k+1)$ is the amount of payment in year k, here an amount that increases by £1 each year.*
- *The discount factor v^{k+1} tells us when the payment has occurred: here, at the end of year of death,*
- *The probability ${}_{k|}q_x$ tells us the payment occurs only if (x) dies between ages k and $k+1$, i.e. a death benefit,*
- *The sum starts at $k=0$, which means there is no deferral period.*
- *The sum goes up to $k=\infty$, which means it is a whole life insurance.*

Example 4.56 (Term insurance paid at the end of year) *Suppose an n-year term insurance pays a sum insured of £(K_x+1) at the end of year of death, if death occurs within n years. The present value random variable, Z, of the insurance is*

$$Z := v^{K_x+1}\,(K_x+1)\,\mathbb{1}_{[T_x\le n]}.$$

The expected present value of the benefit is

$$\boxed{(IA)^{1}_{x:\overline{n}|} := \sum_{k=0}^{n-1} (k+1)\; v^{k+1}\;{}_{k|}q_x.}$$

Note that

$$\lim_{n\to\infty} (IA)^1_{x:\overline{n}|} = (IA)_x.$$

If you wish to look up the EPV of an increasing insurance from a life table, the values are available only for $\{(IA)_x; x = 18, 19, \ldots\}$. In order to calculate $(IA)^1_{x:\overline{n}|}$, we write it in terms of EPVs we can find in the life table.

Exercise 4.57 *Show that*

$$\boxed{(IA)^1_{x:\overline{n}|} = (IA)_x - v^n\, {}_np_x\, ((IA)_{x+n} + nA_{x+n})\,.}$$

It may help to sketch the annual death payments, to understand the formula.

In the third line, changing the index of summation from k to $r = k - n$,

$$\begin{aligned}
(IA)^1_{x:\overline{n}|} = \sum_{k=0}^{n-1} (k+1)\; v^{k+1}\, {}_{k|}q_x &= \sum_{k=0}^{\infty} (k+1)\; v^{k+1}\, {}_{k|}q_x - \sum_{k=n}^{\infty} (k+1)\; v^{k+1}\, {}_{k|}q_x \\
&= (IA)_x - \sum_{k=n}^{\infty} (k+1)\; v^{k+1}\, {}_{k|}q_x \\
&= (IA)_x - \sum_{r=0}^{\infty} (r+n+1)\; v^{r+n+1}\, {}_{r+n|}q_x \\
&= (IA)_x - v^n\, {}_np_x \sum_{r=0}^{\infty} (r+n+1)\; v^{r+1}\, {}_{r|}q_{x+n} \\
&= (IA)_x - v^n\, {}_np_x \left(\sum_{r=0}^{\infty} (r+1)\; v^{r+1}\, {}_{r|}q_{x+n} + \sum_{r=0}^{\infty} n\, v^{r+1}\, {}_{r|}q_{x+n} \right) \\
&= (IA)_x - v^n\, {}_np_x\, ((IA)_{x+n} + nA_{x+n})\,.
\end{aligned}$$

Exercise 4.58 *From the Formulae and Tables for Actuarial Examinations 2025, at an effective interest rate of 4% per annum and on the mortality basis TMNL16,*

$(IA)_{30} = 6.06898$, $(IA)_{65} = 8.68634$, $A_{65} = 0.42195$, $\ell_{30} = 9970.2407$, $\ell_{65} = 9525.1197$.

Calculate $(IA)^1_{30:\overline{35}|}$.

With $x = 30$ and $n = 35$,

$$\begin{aligned}
(IA)^1_{30:\overline{35}|} &= (IA)_{30} - v^{35}\, {}_{35}p_{30}\, ((IA)_{65} + 35A_{65}) \\
&= 6.06898 - (1.04)^{-35} \times \frac{9525.1197}{9970.2407} \times (8.68634 + 35(0.42195)) \\
&= 6.06898 - (0.242102) \times (8.68634 + 35(0.42195)) \\
&= 6.06898 - (0.242102) \times (23.45459) \\
&= 0.39058.
\end{aligned}$$

Lastly, the increasing endowment insurance has a survival benefit that also increases annually.

Example 4.59 (Endowment insurance paid at the end of year) *Suppose an n-year endowment insurance pays a sum insured of $£(K_x+1)$ at the end of year of death, if death occurs within n years. If the policyholder survives n years, then a survival benefit of $£n$ is paid immediately. The present value random variable, Z, of the insurance is*

$$Z := v^{K_x+1}\,(K_x+1)\,\mathbb{1}_{[T_x\le n]} + v^n\,n\,\mathbb{1}_{[T_x>n]}.$$

The expected present value of the benefit is

$$\boxed{(IA)_{x:\overline{n}|} := (IA)^1_{x:\overline{n}|} + nv^n\,{}_np_x.}$$

Benefits increasing annually and paid immediately

Now turn to insurances where the death benefit increases by £1 each year, but it is paid immediately upon death.

Example 4.60 (Whole life insurance increasing annually and paid immediately) *Suppose a whole life insurance pays a sum insured of $£(K_x+1)$ immediately upon death. The present value random variable, Z, of the insurance is*

$$Z := v^{T_x}\,(K_x+1) = v^{T_x}\,\left(\lfloor T_x\rfloor + 1\right).$$

The expected present value of the benefit is

$$(I\bar{A})_x := \mathbb{E}\left(v^{T_x}\,\left(\lfloor T_x\rfloor+1\right)\right) = \int_0^\infty \left(\lfloor t\rfloor+1\right)\,v^t\,{}_tp_x\,\mu_{x+t}\,dt.$$

Partitioning the integral into years, the benefit payment is $£(k+1)$ in year k, for $k = 0,1,2,\ldots$. Hence

$$(I\bar{A})_x = \sum_{k=0}^{\infty}(k+1)\int_k^{k+1} v^t\,{}_tp_x\,\mu_{x+t}\,dt.$$

As $\int_k^{k+1} v^t\,{}_tp_x\,\mu_{x+t}\,dt = {}_{k|}\bar{A}^1_{x:\overline{1}|}$, we get

$$\boxed{(I\bar{A})_x = \sum_{k=0}^{\infty}(k+1)\,{}_{k|}\bar{A}^1_{x:\overline{1}|}.}$$

The expression for $(I\bar{A})_x$ is useful, because it can be easily approximated, allowing us to evaluate it using a life table.

Exercise 4.61 *Show that, under the claims acceleration approach,*

$$\boxed{(I\bar{A})_x = (1+i)^{0.5}\,(IA)_x}$$

and, assuming UDD between integer ages,

$$\boxed{(I\bar{A})_x = \frac{i}{\delta}\,(IA)_x.}$$

Remember that $A^1_{x:\overline{1}|} = v\,q_x$ and ${}_{k|}A^1_{x:\overline{1}|} = v^k\,{}_kp_x\,A^{\ 1}_{x+k:\overline{1}|} = v^{k+1}\,{}_{k|}q_x$.

Under the claims acceleration approach, $\bar{A}^{1}_{x:\overline{1}|} = (1+i)^{0.5}\, A^{1}_{x:\overline{1}|}$ which means

$$ {}_{k|}\bar{A}^{1}_{x:\overline{1}|} = v^{k}\, {}_{k}p_x\, \bar{A}^{\ 1}_{x+k:\overline{1}|} \overset{\text{Claims acc.}}{=} (1+i)^{0.5}\, v^{k}\, {}_{k}p_x\, A^{1}_{x:\overline{1}|} = (1+i)^{0.5}\, {}_{k|}A^{1}_{x:\overline{1}|}. $$

Hence

$$ \begin{aligned} (I\bar{A})_x = \sum_{k=0}^{\infty}(k+1)\, {}_{k|}\bar{A}^{1}_{x:\overline{1}|} \overset{\text{Claims acc.}}{=} (1+i)^{0.5} \sum_{k=0}^{\infty}(k+1)\, {}_{k|}A^{1}_{x:\overline{1}|} &= (1+i)^{0.5} \sum_{k=0}^{\infty}(k+1)\, v^{k+1}\, {}_{k|}q_x \\ &= (1+i)^{0.5}\, (IA)_x. \end{aligned} $$

Assuming UDD between integer ages, $\bar{A}^{1}_{x:\overline{1}|} = \frac{i}{\delta}\, A^{1}_{x:\overline{1}|}$. As this is only a constant factor difference to the claims acceleration approach, exactly the same arguments can be applied to get

$$ (I\bar{A})_x = \frac{i}{\delta}\, (IA)_x. $$

Exercise 4.62 *From the Formulae and Tables for Actuarial Examinations 2025, at an effective interest rate of 4% per annum and using the life table TMNL16,* $(IA)_{30} = 6.06898$. *Calculate* $(I\bar{A})_{30}$.

We can use either claims acceleration or UDD, since it is not specified in the question. Both possibilities are shown here, for you to check your answer.

Using claims acceleration

$$ (I\bar{A})_{30} = (1+i)^{0.5}\, (IA)_{30} = (1.04)^{0.5} \times 6.06898 = 6.18917. $$

Assuming UDD between integer ages,

$$ (I\bar{A})_{30} = \frac{i}{\delta}\, (IA)_{30} = \frac{0.04}{\ln(1.04)} \times 6.06898 = 6.18957. $$

Example 4.63 (Term insurance increasing annually and paid immediately) *Suppose an* n*-year term insurance pays a sum insured of* $£(K_x+1)$ *immediately upon death, if death occurs within* n *years. The present value random variable,* Z*, of the insurance is*

$$ Z := v^{T_x}\, (K_x+1)\, \mathbb{1}_{[T_x \le n]}. $$

Following the same argument as for the whole life insurance, the expected present value of the benefit is

$$ \boxed{(I\bar{A})^{1}_{x:\overline{n}|} = \sum_{k=0}^{n-1}(k+1)\, {}_{k|}\bar{A}^{1}_{x:\overline{1}|}.} $$

Exercise 4.64 *Show that, under the claims acceleration approach,*

$$\boxed{(I\bar{A})^{1}_{x:\overline{n}|} = (1+i)^{0.5}\,(IA)^{1}_{x:\overline{n}|}}$$

and, assuming UDD between integer ages,

$$\boxed{(I\bar{A})^{1}_{x:\overline{n}|} = \frac{i}{\delta}\,(IA)^{1}_{x:\overline{n}|}.}$$

Use the same argument as used above, for the whole life insurance.

Exercise 4.65 *From Exercise 4.62,* $(IA)^{1}_{30:\overline{35}|} = 0.39058$. *On the same basis, i.e. the life table TMNL16 and an interest rate of 4% per annum, calculate* $(I\bar{A})^{1}_{30:\overline{35}|}$.

We can use either claims acceleration or UDD, since it is not specified in the question. Both possibilities are shown here, for you to check your answer.

Using claims acceleration

$$(I\bar{A})^{1}_{30:\overline{35}|} = (1+i)^{0.5}\,(IA)^{1}_{30:\overline{35}|} = (1.04)^{0.5}\,0.39058 = 0.39832.$$

Assuming UDD between integer ages,

$$(I\bar{A})^{1}_{30:\overline{35}|} = \frac{i}{\delta}\,(IA)^{1}_{30:\overline{35}|} = \frac{0.04}{\ln(1.04)}\,0.39058 = 0.39834.$$

Example 4.66 (Endowment insurance increasing annually and paid immediately) *Suppose an n-year endowment insurance pays a sum insured of* $£(K_x+1)$ *immediately upon death, if death occurs within n years. If the policyholder survives n years, then a survival benefit of* $£n$ *is paid immediately. The present value random variable, Z, of the insurance is*

$$Z := v^{T_x}\,(K_x+1)\,\mathbb{1}_{[T_x\le n]} + v^{n}\,n\,\mathbb{1}_{[T_x>n]}.$$

The expected present value of the benefit is

$$\boxed{(I\bar{A})_{x:\overline{n}|} = (I\bar{A})^{1}_{x:\overline{n}|} + nv^{n}\,{}_{n}p_x.}$$

Exercise 4.67 *Show that, under the claims acceleration approach,*

$$\boxed{(I\bar{A})_{x:\overline{n}|} = (1+i)^{0.5}\,(IA)^{1}_{x:\overline{n}|} + nv^{n}\,{}_{n}p_x.}$$

and, assuming UDD between integer ages,

$$\boxed{(I\bar{A})_{x:\overline{n}|} = \frac{i}{\delta}\,(IA)^{1}_{x:\overline{n}|} + nv^{n}\,{}_{n}p_x.}$$

Follows from the previous exercise applied to $(I\bar{A})^{1}_{x:\overline{n}|}$.

Benefits increasing continuously and paid immediately

Example 4.68 (Whole life insurance increasing continuously and paid immediately) *Suppose a whole life insurance pays a sum insured of £T_x immediately upon death. The present value random variable, Z, of the insurance is*

$$Z := v^{T_x}\, T_x.$$

The expected present value of the benefit is

$$\boxed{(\bar{I}\bar{A})_x := \int_0^\infty t\, v^t\, {}_tp_x\, \mu_{x+t}\, dt.}$$

Example 4.69 (Term insurance increasing continuously and paid immediately) *Suppose an n-year term insurance pays a sum insured of £T_x immediately upon death, if death occurs within n years. The present value random variable, Z, of the insurance is*

$$Z := v^{T_x}\, T_x\, \mathbb{1}_{[T_x \le n]}.$$

The expected present value of the benefit is

$$\boxed{(\bar{I}\bar{A})^1_{x:\overline{n}|} := \int_0^n t\, v^t\, {}_tp_x\, \mu_{x+t}\, dt.}$$

Note that

$$\lim_{n\to\infty} (\bar{I}\bar{A})^1_{x:\overline{n}|} = (\bar{I}\bar{A})_x.$$

Example 4.70 (Endowment insurance increasing continuously and paid immediately) *Suppose an n-year endowment insurance pays a sum insured of £T_x immediately upon death, if death occurs within n years. If the policyholder survives n years, then a survival benefit of £n is paid immediately. The present value random variable, Z, of the insurance is*

$$Z := v^{T_x}\, T_x\, \mathbb{1}_{[T_x \le n]} + v^n\, n\, \mathbb{1}_{[T_x > n]}.$$

The expected present value of the benefit is

$$\boxed{(\bar{I}\bar{A})_{x:\overline{n}|} := (\bar{I}\bar{A})^1_{x:\overline{n}|} + n v^n\, {}_np_x.}$$

Exercises on increasing insurances

Exercise 4.71 *Explain the following result intuitively:*

$$(IA)^1_{x:\overline{n}|} = (n+1)A^1_{x:\overline{n}|} - \sum_{k=1}^{n} A^1_{x:\overline{k}|}.$$

It may help you to sketch the benefit amounts in each year.

- The EPV $(n+1)A^1_{x:\overline{n}|}$ represents a death benefit of £$(n+1)$ in year r, for $r = 1, 2, \ldots, n-1$, payable at the end of year of death.
- The sum of EPVs $\sum_{k=1}^{n} A^1_{x:\overline{k}|}$ represents a death benefit of £n in year 1, £$(n-1)$ in year 2, ..., £1 in year n.
- Subtracting the sum $\sum_{k=1}^{n} A^1_{x:\overline{k}|}$ from $(n+1)A^1_{x:\overline{n}|}$, gives a death benefit of £1 in year 1, £2 in year 2, ..., £n in year n.
- This payment stream is the same one represented by the EPV $(IA)^1_{x:\overline{n}|}$.

Exercise 4.72 *Explain why, with a positive rate of interest, you would expect:*

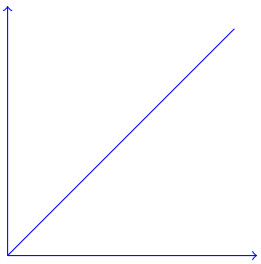
$$(I\bar{A})^1_{x:\overline{n}|} > (\bar{I}\bar{A})^1_{x:\overline{n}|}.$$

The timing of the death benefit payment and probability its occurence is the same under both insurances. However the amount of death benefit is different. Let $k \in \{0, 1, \ldots, n-1\}$ and suppose that death occurs at time $T_x \in [k, k+1)$.

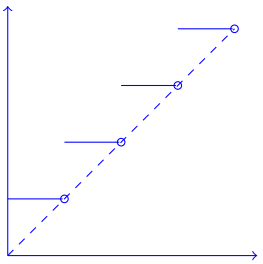
- Under the EPV $(I\bar{A})^1_{x:\overline{n}|}$, the amount of death benefit is £$(k+1)$.
- Under the EPV $(\bar{I}\bar{A})^1_{x:\overline{n}|}$, the amount of death benefit is £$T_x \leq k+1$.

As the death benefit in the insurance with EPV $(\bar{I}\bar{A})^1_{x:\overline{n}|}$ is less than under $(I\bar{A})^1_{x:\overline{n}|}$, then - with the timing and probabilities identical in both EPVs - it follows that $(\bar{I}\bar{A})^1_{x:\overline{n}|} < (I\bar{A})^1_{x:\overline{n}|}$.

$(\bar{I}\bar{A})^1_{x:\overline{n}|}$



$(I\bar{A})^1_{x:\overline{n}|}$



Geometrically increasing benefits

Geometrically increasing benefits represent a stylisation of long-term insurance contracts called *with-profits contracts* in the UK or *participating policies* in North America. These are contracts in which the benefits increase roughly in line with the investment returns earned on the assets, with the actual increase awarded each year decided by the insurance company. They are studied in Chapter 6.

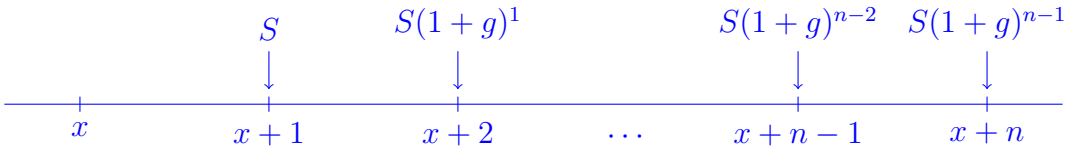
Exercise 4.73 (Geometrically increasing death benefits) *Fix a constant $g > -1$. An n-year term insurance pays the sum insured at the end of the year of death of (x). If (x) dies between ages $x+k$ and $x+k+1$, the amount of sum insured is £$S(1+g)^k$, for $k = 0, 1, \ldots, n-1$.*

Show that the EPV of this benefit can be written

$$\frac{S}{1+g}\, A^1_{x:\overline{n}|},$$

where $A^1_{x:\overline{n}|}$ is calculated using an interest rate

$$j := \frac{1+i}{1+g} - 1.$$



The EPV is

$$\begin{aligned}\sum_{k=0}^{n-1} S(1+g)^k\, v^{k+1}\, {}_{k|}q_x &= \frac{S}{1+g} \sum_{k=0}^{n-1} \left(\frac{1+g}{1+i}\right)^{k+1} {}_{k|}q_x \\ &= \frac{S}{1+g}\, A^1_{x:\overline{n}|}, \qquad \text{at interest rate } j := \frac{1+i}{1+g} - 1.\end{aligned}$$